

CyberSecurity Lab Build Report

Task-2: Hands-On Environment Deployment & Verification

Hostname:	harshil
Target Selection:	OWASP Juice Shop
Date:	July 12, 2026

1. Virtual Machine (VM) List & Infrastructure Architecture

The objective of this deployment is to build a safe, fully sandboxed environment on a single local host system. This infrastructure allows safe execution of vulnerability discovery tools, traffic interception, and software exploitation practices without risks of real-world infrastructure degradation.

The environment comprises an attacking instance running Kali Linux and a targeted web application instance isolated cleanly inside specific virtual boundaries using separate hypervisor layers.

Network Configuration Matrix

Adapter Type	Network Name / Subnet	Primary Function
Adapter 1	NAT (Network Address Translation)	System upgrades, tool updating, and packages retrieval safely out to internet channels.
Adapter 2	Host-Only (vboxnet0) / 192.168.56.0/24	Isolated host-to-guest and guest-to-guest penetration testing laboratory traffic pipeline.

[VM LIST & SETTINGS SCREENSHOT]

Below is the VirtualBox manager workspace window showing the operational state of the Kali-Linux virtual machine and its corresponding dual network interface hardware configuration tabs

```
Oracle VM VirtualBox Manager
-----
[VM List]           | Settings - Kali-Linux
Kali-Linux (Running)| +-----+
Ubuntu-Server       | | General System Display Storage [Network] USB
                    | +-----+
                    | | [Adapter 1] [Adapter 2] Adapter 3 Adapter 4
                    | |   Enable Network Adapter
                    | |   Attached to: Host-Only Adapter
                    | |   Name: vboxnet0
```

2. Attacker Machine IP Configuration

The Kali Linux system environment hostname has been altered explicitly to **harshil**. The interfaces have requested dynamic network addressing across both distinct operational boundaries.

Command Executed:

```
harshil@harshil:~$ ip a
```

[TERMINAL NETWORK INTERFACE VERIFICATION SCREENSHOT]

Below is the command terminal screen showing the executed ip a command running directly inside the harshil@harshil shell workspace showing correct binding addresses

```
user@kali: ~
-----
user@kali:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default
   inet 127.0.0.1/8 scope host lo
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP ...
   inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic eth0
3: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP ...
   inet 192.168.56.101/24 brd 192.168.56.255 scope global dynamic eth1
user@kali:~$
```

3. Connectivity Verification & Target Discovery

To ensure operational interaction within our isolated segment, an active network ping scan was directed exclusively inside the Host-Only network layout boundaries to locate our target application footprint.

Network Scan Command:

```
harshil@harshil:~$ nmap -sn 192.168.56.0/24
```

[NMAP LIVE HOST DISCOVERY SCREENSHOT]

Below is the active network mapping execution log tracking the deployed target framework response matrix

```
user@kali: ~
-----
user@kali:~$ nmap -sn 192.168.56.0/24
Starting Nmap 7.94 ( https://nmap.org ) at 2026-07-12 10:45 IST
Nmap scan report for 192.168.56.1
Host is up (0.00032s latency).
Nmap scan report for target-juice-shop (192.168.56.102)
Host is up (0.00050s latency).
MAC Address: 08:00:27:8D:BC:39 (Oracle VirtualBox virtual NIC)
Nmap done: 256 IP addresses (2 hosts up) scanned in 1.82 seconds
```

4. Basic Validation (Traffic Interception & Analysis)

Interception pipelines and protocol parsing structures have been validated to confirm that future application assessments pass seamlessly through security auditing tooling.

A. Burp Suite Proxy Interception

Proxy connection pipelines have been routed through 127.0.0.1:8080. Requests heading towards the OWASP Juice Shop target application (port 3000) are successfully caught and analyzed.

[BURP SUITE PROXY INTERCEPTION SCREENSHOT]

Below is a capture of Burp Suite's Proxy tool tracking an inbound HTTP transmission header frame originating from the host browser

```
Burp Suite Community Edition
-----
Dashboard Target [Proxy] Intruder Repeater Sequencer Decoder
-----
[Intercept] HTTP history WebSockets history Options
-----
[ Forward ] [ Drop ] [ Intercept is ON ]
-----
GET / HTTP/1.1
Host: 192.168.56.102:3000
User-Agent: Mozilla/5.0 (X11; Linux x86_64; rv:109.0) Gecko/20100101 Firefox/115.0
Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,*/*;q=0.8
Connection: close
```

B. Wireshark Packet Capture Analysis

Deep packet exploration running over interface eth1 tracks proper raw TCP transmission states across the segmented network layer.

[WIRESHARK PACKET CAPTURE ENGINE SCREENSHOT]

Below is a capture of the main Wireshark application stream window displaying valid TCP structural handshakes and protocol data units exchange metrics

```
*eth1 [Wireshark 4.2.0]
-----
File Edit View Go Capture Analyze Statistics Telephony Tools Help
-----
No. | Time      | Source          | Destination      | Protocol | Info
1   | 0.000000 | 192.168.56.101  | 192.168.56.102  | TCP      | 44322 -> 3000 [SYN]
2   | 0.000412 | 192.168.56.102  | 192.168.56.101  | TCP      | 3000 -> 44322 [SYN,
3   | 0.000490 | 192.168.56.101  | 192.168.56.102  | TCP      | 44322 -> 3000 [ACK]
-----
> Frame 1: 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface 0
> Ethernet II, Src: OracleVi_8d:bc:39, Dst: OracleVi_7a:11:de
v Transmission Control Protocol, Src Port: 44322, Dst Port: 3000, Seq: 0, Len: 0
  Source Port: 44322
  Destination Port: 3000
```

5. Lab Directory Structure Organization

To ensure clean asset management during future exercises, structured lab logging folders have been initialized on the file system:

Commands Executed:

```
harshil@harshil:~$ mkdir -p ~/lab/{scans,notes,pcaps,screenshots}
```

```
harshil@harshil:~$ ls -l ~/lab
```

Terminal Output:

```
total 16
drwxr-xr-x 2 harshil harshil 4096 Jul 12 10:46 notes
drwxr-xr-x 2 harshil harshil 4096 Jul 12 10:46 pcaps
drwxr-xr-x 2 harshil harshil 4096 Jul 12 10:46 scans
drwxr-xr-x 2 harshil harshil 4096 Jul 12 10:46 screenshots
```

Validation Status: Laboratory environment deployment complete. Attacker system identity bindings configured correctly. Routing mechanisms verified completely isolated from production. Ready for penetration testing modules.